

epMotion Equimolar Pooling Helper

Detailed User Guide - for molarity checks, quick manual pooling, and full epMotion worklist generation.

The tool runs locally in the browser. It converts library concentration and size into molarity, calculates equal molecular contributions, checks pipetting limits, previews the source rack, and exports bench-ready files.

1. Choose the right workflow

Workflow	Use when	Minimum required fields	Output
Calculate Molarities	You only want to convert library measurements into nM before deciding how to pool.	For each sample: concentration reads + size bp, or direct nM.	Molarity table. No pooling volumes.
Quick Pool	You want a fast manual pooling table, usually for bench calculation or ONT fmol-style pooling.	Sample molarity inputs + fmol per sample or total fmol pool.	Manual transfer volumes and quick pool CSV.
Generate Worklists	You need robot-ready epMotion worklists with DNA and buffer transfers.	Sample molarity inputs + pool mode + total final volume + layout.	DNA.csv, Buffer.csv, worksheet, inputs file, optional pre-dilution plan.

2. Pooling section vs Sample section

Section	Meaning	Examples
Sample section	Describes what you currently have. These are measured values from the library QC stage. It can be used alone for Calculate Molarities and Quick Pool.	Sample name, source well, Qubit reads, average fragment size, direct nM.
Create Pool Parameters	Describes the final pool you want to create. Required only for full pool/worklist generation.	Target nM or total fmol, total final volume, one-tube or normalize layout.
Optional extra pooling parameters	Fine-tunes robot handling. These fields have defaults and are not required for normal use.	Minimum transfer volume, rounding step, excess %, rack numbers, pool well, low-volume handling, tip policy.

3. Sample entry rules

Each sample needs either **ng/μL + bp** or **direct nM**. For replicate Qubit readings, enter values separated by commas, semicolons, or spaces, for example: **23.1, 23.8, 24.0**. The tool calculates a row-level average internally and uses that average for molarity. The separate sample summary below the table displays the overall average concentration across entered mass-mode samples.

Field	Required?	Purpose
Source well	Recommended	Maps the sample onto the live source rack. Auto-filled if blank.
Name / number	Recommended	Appears in tables, exports, and the plate map.
Input mode	Yes	Choose ng/μL + bp or direct nM.
Conc reads ng/μL	Yes for mass mode	Qubit or fluorometric concentration readings. Multiple reads are averaged.
Size bp	Yes for mass mode	Average fragment size from TapeStation, Bioanalyzer, Fragment Analyzer, etc.
Direct nM	Yes for direct nM mode	Use when molarity was already measured or provided.

4. Button reference

Button	What it does
Calculate Molarities	Uses only the Sample section and reports molarity in nM. It does not create a pool.
Quick Pool	Creates a manual pooling table using equal fmol per sample or a total fmol split equally across samples.
Generate Worklists	Runs the full epMotion calculation and creates worklist files.
Import instrument CSV/TXT	Imports sample names, wells, concentrations, sizes, or direct nM values when the file has recognizable columns.
Export contents CSV/TXT	Exports the current inputs for review or record keeping.
Save Run / Open Run	Saves or reloads the complete browser session as JSON. Use this for reproducibility.
Blank CSV template	Downloads a template showing the expected import format.
New / clear	Clears the current session. Use carefully.
User Guide	Downloads this detailed PDF guide directly.

5. Outputs and interpretation

Output	Meaning	Use
DNA.csv	epMotion DNA/library transfer worklist.	Load first for robotic DNA transfers.
Buffer.csv	epMotion buffer transfer worklist.	Load after DNA.csv.
Pooling worksheet	Human-readable calculation table with molarity, raw/rounded volumes, and fmol contribution.	Review, print, archive with run records.
Inputs file	Original sample and run settings.	Reproducibility and troubleshooting.
Pre-dilution plan	Generated only when enabled and required for concentrated samples below minimum transfer volume.	Perform before running final pooling worklist.

6. Practical checks before running epMotion

Confirm that the live source rack plate map matches the physical rack. Check that all required fields marked with an asterisk are filled. Review low-volume warnings. Transfers below the validated minimum should be handled by increasing the final pool volume or pre-diluting concentrated libraries. Only TS_50 and TS_300 tips are assumed by this tool.

Equimolar pooling means each library contributes the same number of molecules, not the same volume. High-molarity libraries require smaller volumes; low-molarity libraries require larger volumes. Always follow your laboratory SOP when selecting target nM, total fmol, final volume, and acceptance thresholds.